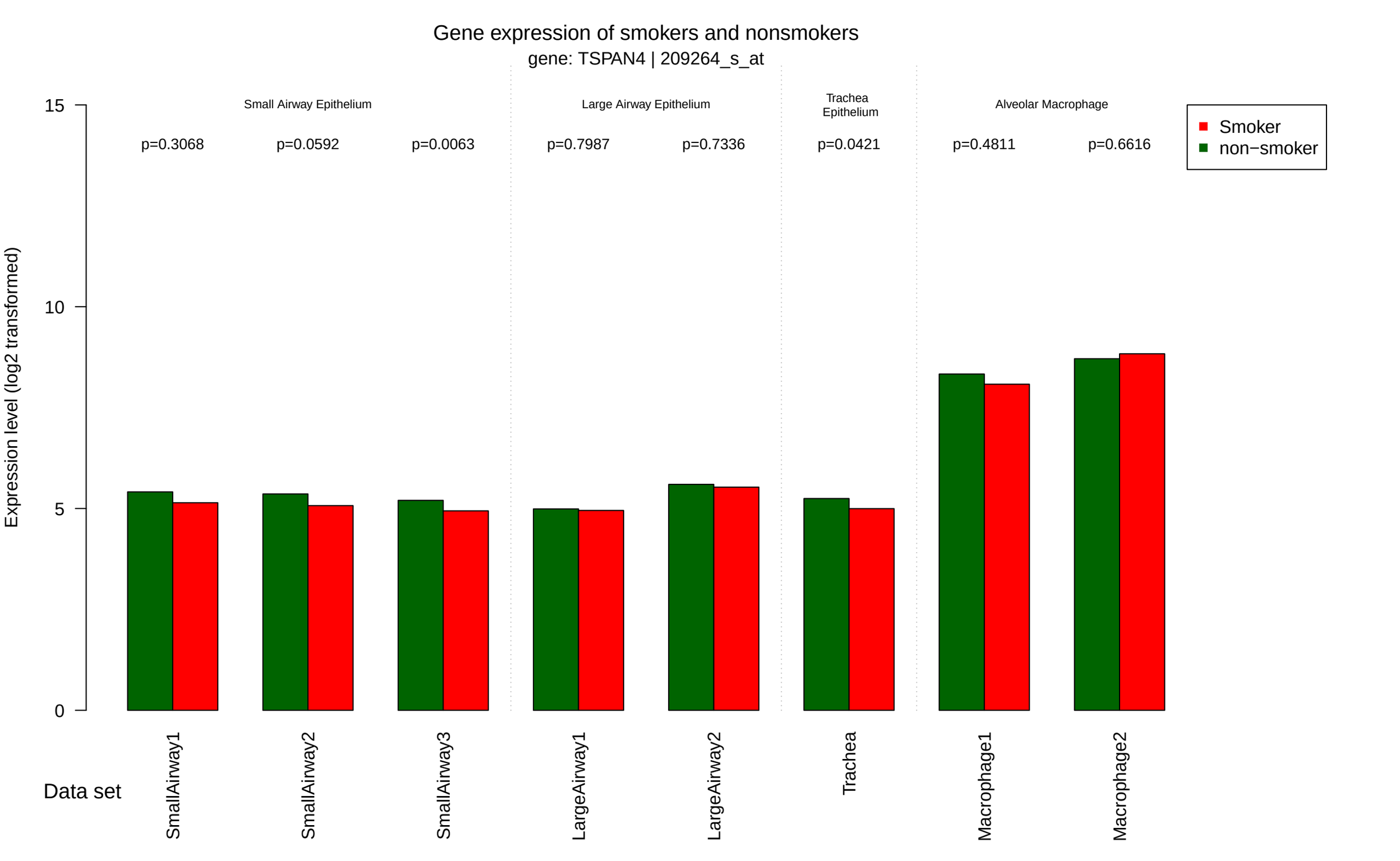
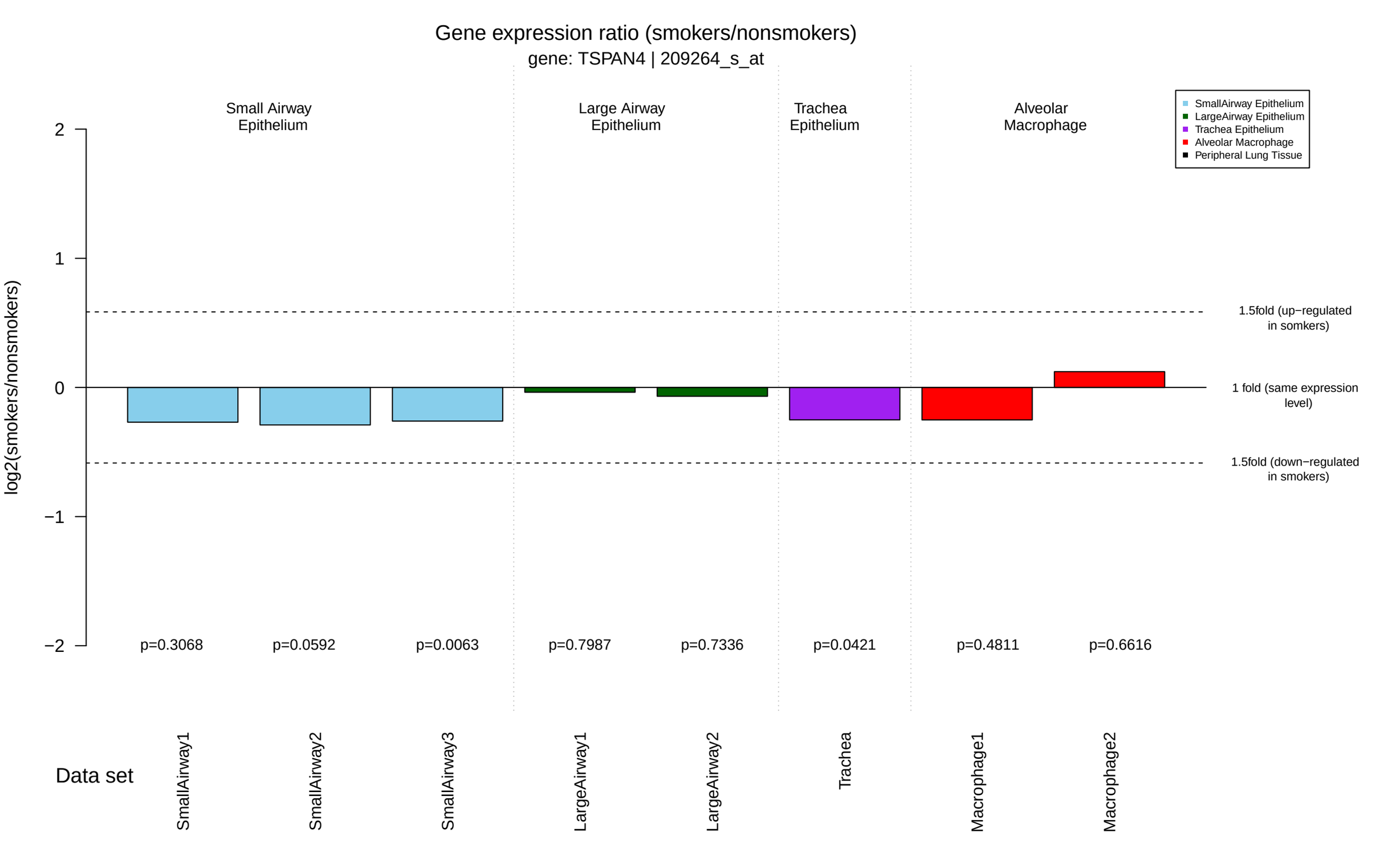
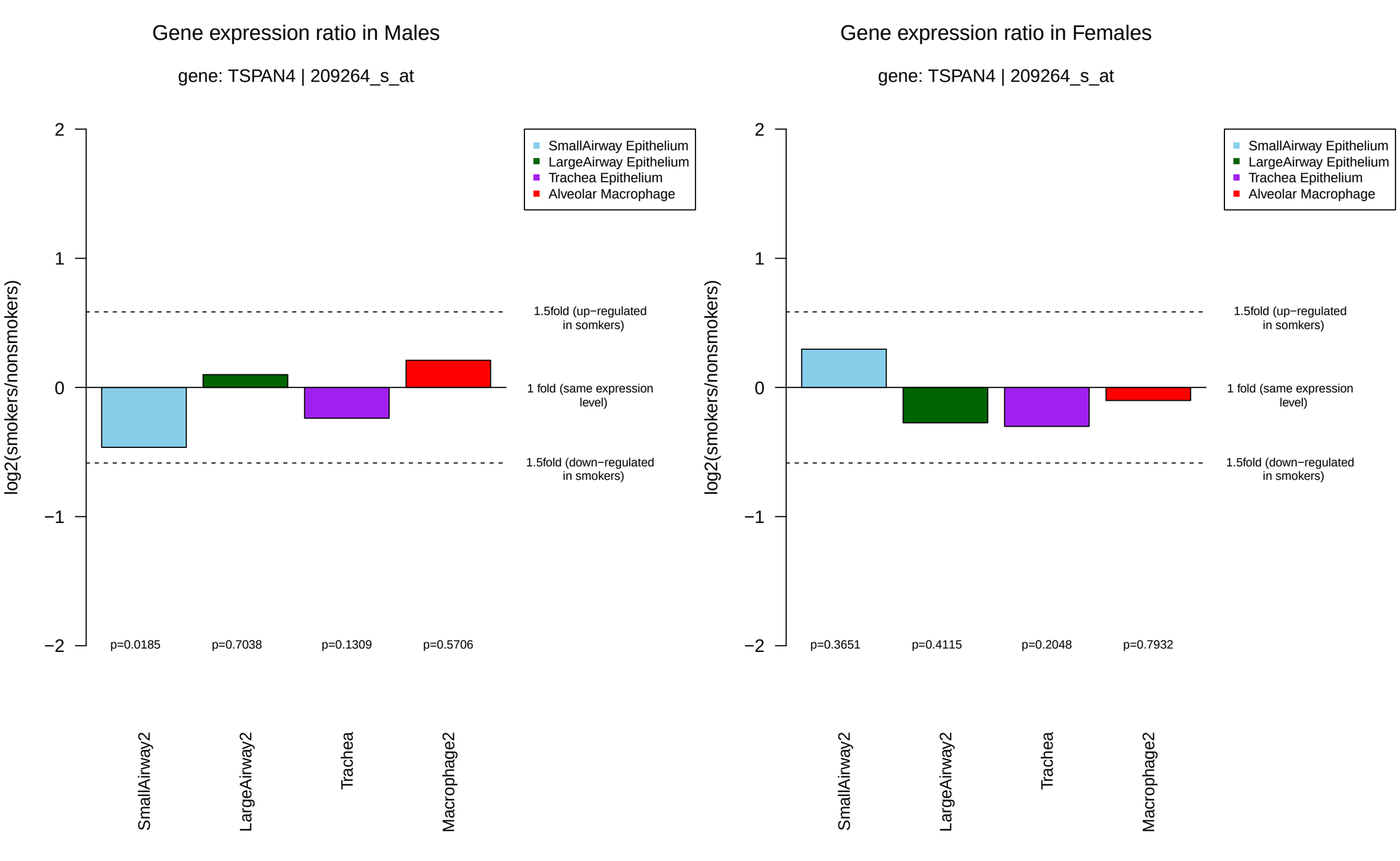
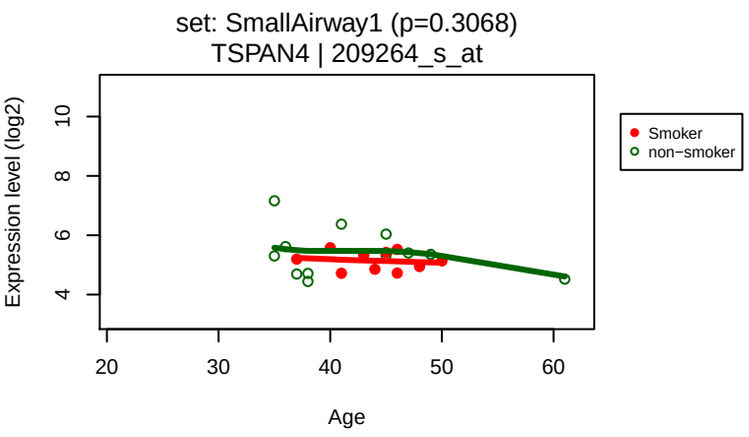
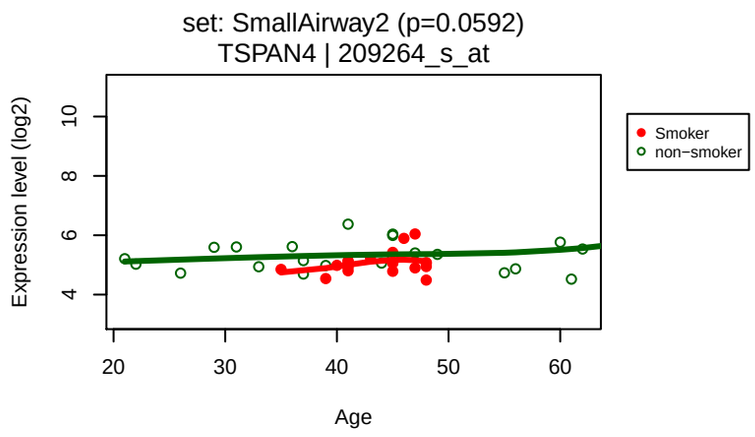
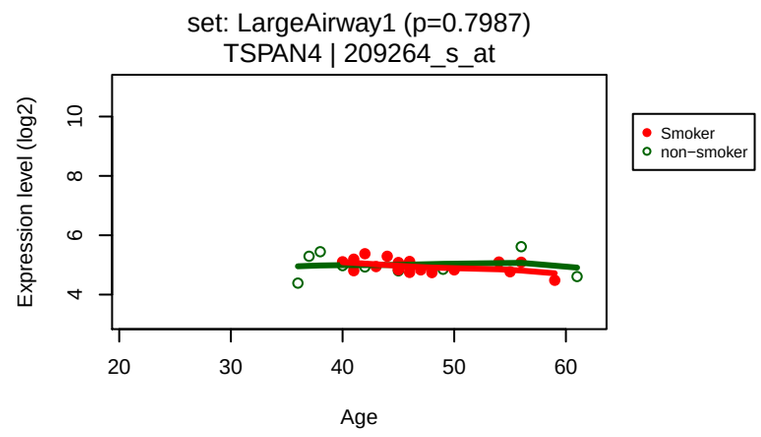

Gene expression of smokers and nonsmokers
gene: TSPAN4 | 209264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3068
p=0.0592
p=0.0063
p=0.7987
p=0.7336
p=0.0421
p=0.4811
p=0.6616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

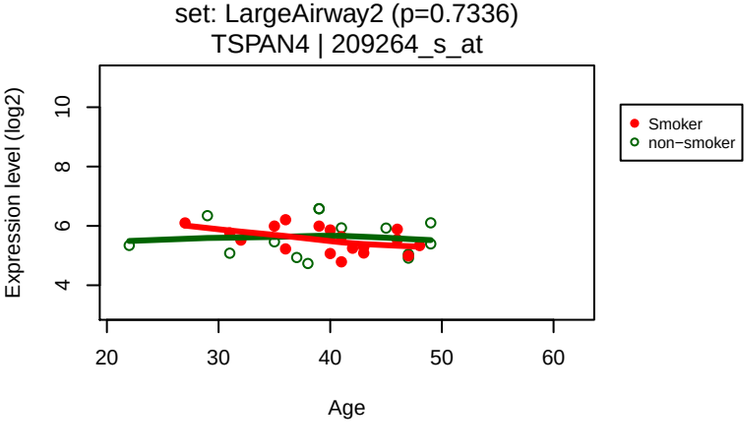

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN4 | 209264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3068
p=0.0592
p=0.0063
p=0.7987
p=0.7336
p=0.0421
p=0.4811
p=0.6616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

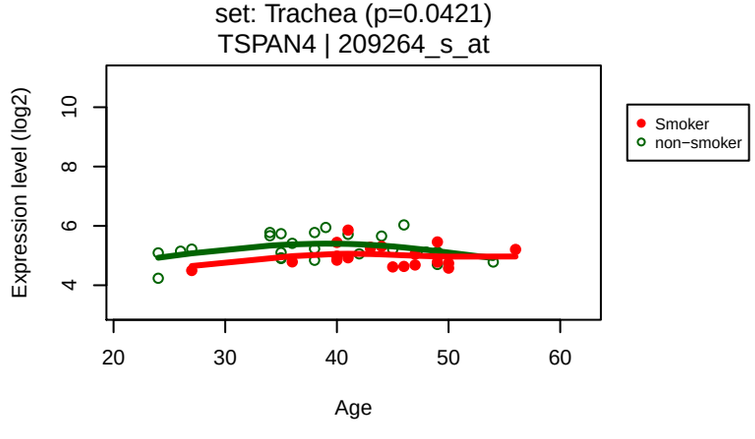

Gene expression ratio in Males
gene: TSPAN4 | 209264_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0185
p=0.7038
p=0.1309
p=0.5706
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN4 | 209264_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3651
p=0.4115
p=0.2048
p=0.7932
SmallAirway2
LargeAirway2
Trachea
Macrophage2

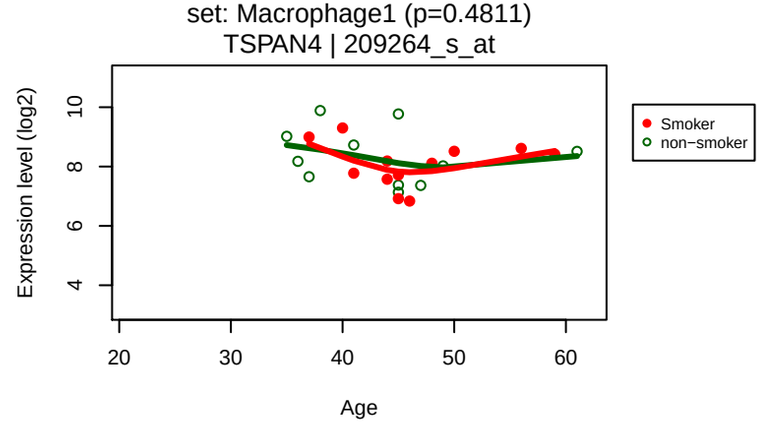

set: SmallAirway1 (p=0.3068)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker

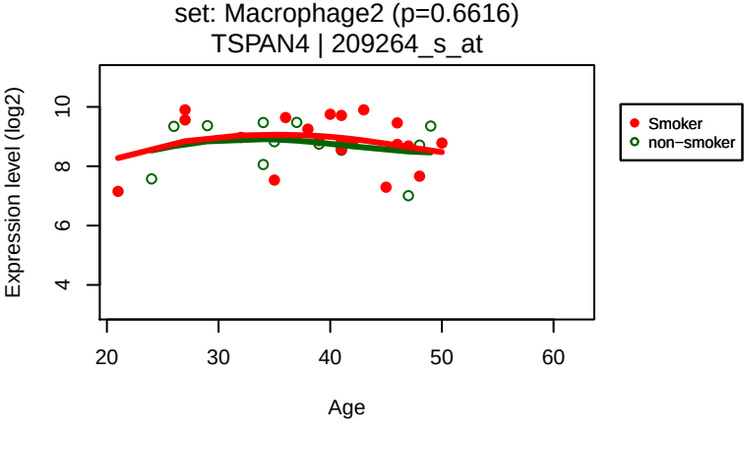

set: SmallAirway2 (p=0.0592)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker

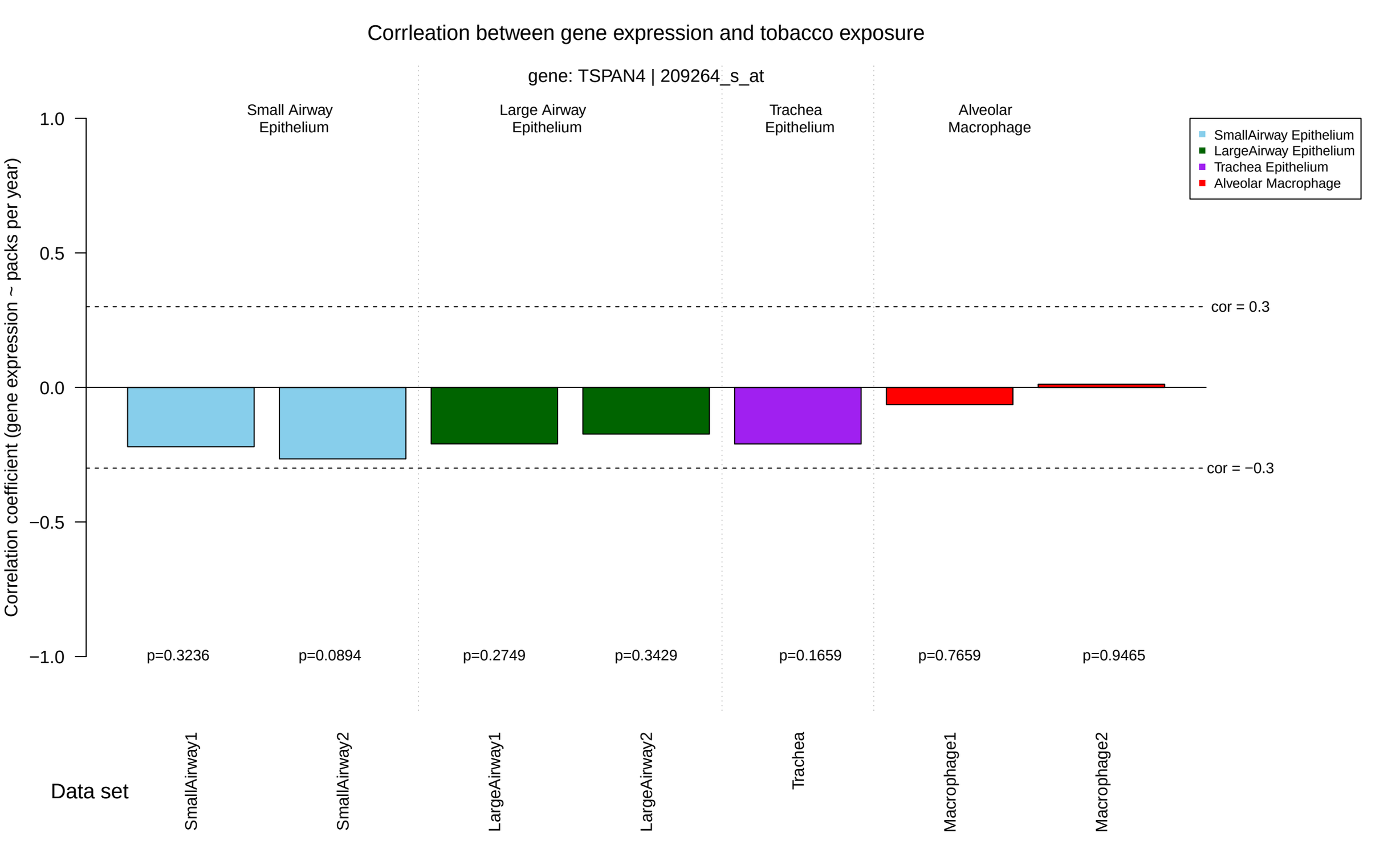

set: LargeAirway1 (p=0.7987)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7336)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0421)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4811)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6616)
TSPAN4 | 209264_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN4 | 209264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3236
p=0.0894
p=0.2749
p=0.3429
p=0.1659
p=0.7659
p=0.9465
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2